

Homework 1

Instructions

Complete the following problems and submit to Gradescope your work as a file named `hw1.Rmd`.

A file with the wrong name or a bug will prevent the autograder from being able to run your submission. Check that your filename is correct, and check for bugs clearing your memory and then running your file from start to finish. It's also a good idea to check that your file "knits."

Do not use any loops!

Problem 1

part a

Create a **vector** called `p1a` that stores all integers from 513 to 182371 counting by 9.

Hint:

```
tail(p1a)
```

```
## [1] 182322 182331 182340 182349 182358 182367
```

part b

Create a **vector** called `p1b` that stores the string `"1st"` repeated 921390 times.

Hint:

```
head(p1b)
```

```
## [1] "1st" "1st" "1st" "1st" "1st" "1st"
```

```
length(p1b)
```

```
## [1] 921390
```

part c

Create a **vector** called `p1c` that stores the all consecutive integers from 37 to 76876

Hint:

```
head(p1c)
```

```
## [1] 37 38 39 40 41 42
```

```
length(p1c)
```

```
## [1] 76840
```

part d

Create a **vector** called **p1d** that stores the the phrases "1st", "2nd", and "3rd" each repeated 928908 times.

```
head(p1d)
```

```
## [1] "1st" "1st" "1st" "1st" "1st" "1st"
```

```
length(p1d)
```

```
## [1] 2786724
```

part e

Create a **vector** called **p1e** that stores the integers starting at 155 and counting back by 4 until you get to -13875.

```
tail(p1e)
```

```
## [1] -13853 -13857 -13861 -13865 -13869 -13873
```

```
length(p1e)
```

```
## [1] 3508
```

part f

Create a **vector** called **p1f** that looks like this:

```
## [1] 50 60 70 80 90 100 95 90 85 80 75
```

part g

Create a **vector** called **p1g** that looks like this:

```
## [1] 20 20 20 20 30 30 30 40 40 50
```

part h

Create a **vector** called **p1h** that looks like this:

```
## [1] 2 1 0 2 1 0 2 1 0 2 1 0 2 1 0
```

part i

Create a **vector** called **p1i** that looks like this:

```
## [1] 124 116 108 100 92 84 76 68 63 58 53 48 43 38
```

part j

Create a **vector** called **p1j** that looks like this:

```
## [1] "3rd" "2nd" "1st" "3rd" "2nd" "1st"
```

Problem 2

The finishing times of runners in the New York City 10K are Normally distributed with a mean of 63 minutes and a standard deviation of 12 minutes.

a

The fastest road 10K run to date was finished in 29.43 minutes. What is the probability that this time is beaten in this race? Assign your answer to **p2a**.

b

A runner sets a goal to finish in the top 5%. What should the target time for this runner be? Assign your answer to **p2b**.

c

To ensure minimal traffic disruption, the roads will be reopened to traffic exactly 90 minutes after the start of the race. If there are 28,000 runners in the race, how many would you expect to have their races disrupted by the resumption of traffic? Assign your answer to **p2c**.

Problem 3

A federal report found that a lie detector test given to someone telling the truth will suggest that they lied in about 10% of cases. A company asks their 10 job applicants for a new position about thefts from previous employers. All 10 applicants answer honestly.

a

What is the probability that the lie detector indicates that at least one of the applicants is lying? Assign your answer to **p3a**.

b

Assume X is the random variable we are discussing. Create a **vector** called **p3b** that stores the probabilities: $P(X \geq 1), P(X \geq 2), \dots, P(X = 10)$. Round all your numbers to three decimal places.

Problem 4

Write the code to reproduce **and save** the list shown below. When printed, the **list** should match the output below **exactly**. Call the variable **myList**.

```
## $Name
## [1] "Taylor Brown"
##
## $Department
## [1] "Statistics"
##
## $Description
## [1] "He's the best"
##
## $Courses
##      Course1      Course2
## "STAT 3080" "STAT 2559"
##
## $ActiveTeach
## Course1 Course2
##    TRUE  FALSE
##
## $Enr
## $Enr$Course1
## [1] 75 90 90
##
## $Enr$Course2
## [1] 5
##
##
## $Days
## $Days$Course1
##      [,1]      [,2]
## [1,] "Monday" "Wednesday"
## [2,] "Tuesday" "Thursday"
## [3,] "Tuesday" "Thursday"
##
## $Days$Course2
## [1] "Tuesday" "Thursday"
```